AI Equity Basket: Data, Factor Correlation, and Initial Findings

1 Objective

The project builds an interpretable factor framework for an AI-related equity basket covering semiconductors, networking, hyperscalers, data centres, power generation, grid equipment, and nuclear/uranium. The aim is to identify non-redundant drivers of return and risk, then use them for regression and attribution.

2 Data Collection and Cleaning

Daily Yahoo Finance data were collected from 2016-01-01 to 2026-10-01 for stocks, ETFs, indices, yields, commodities, FX, and crypto. The download process is versioned, resumable, and supported by a manifest, per-ticker files, metadata, and quality checks. Of 623 requested series, 622 downloaded successfully.

The cleaning pipeline creates an NYSE-aligned calendar and does not forward-fill prices. It flags missing data, stale prices, invalid OHLC observations, corporate-action inconsistencies, splits, and unusual returns. It produces simple, log, and excess returns, macro levels and changes, dollar volume, eligibility flags, and a complete audit table. The SPY versus S&P 500 validation check passes after excluding SPY ex-dividend dates, with a correlation of 0.9984.

3 Factor Library and Redundancy

The candidate library includes market, rates, credit, volatility, dollar, commodity, style, semiconductor, software, utility, power, grid, and uranium factors. Sector factors are generally expressed as spreads relative to SPY; yield factors are expressed as changes in basis points.

Thirty-four of 38 candidates pass the estimation-window screening. Important redundancy is visible: SEMI and SEMI_SOXX have a market-neutral correlation of 0.978, CYBER and CYBER_HACK have a correlation of 0.914, and SOFT and CLOUD have a correlation of 0.829. The full neutralized factor set is too collinear for direct regression, with a condition number of 269. The primary-factor subset is more usable, with a condition number of 19.7 and maximum VIF of 4.5.

4 Main Findings

The AI basket became more market-sensitive in the holdout period. Its core-basket market beta rose from 1.16 during estimation to 1.74 during holdout.

Semiconductor and software factors explain much of the compute and hyperscaler exposure, although these results must be interpreted carefully because some ETF holdings overlap with basket constituents. The notebook therefore also reports overlap-excluded diagnostics.

The power, grid, and uranium factors provide economically meaningful additional explanatory power. Adding these adjacent factors improves the core basket's out-of-sample explanatory power, particularly for grid and nuclear/uranium exposures. However, factor correlations are

less stable during the holdout and stress periods, supporting the use of rolling or expanding estimation windows.

5 Next Step

The next notebook should use time-ordered cross-validation, penalized regression, and stability selection to choose a compact final factor set. The chosen factors can then be orthogonalized and used for rolling beta estimation, Newey–West inference, and monthly return and risk attribution.

The basket is defined retrospectively, so the results should be interpreted as descriptive factor attribution rather than a fully investable historical backtest.